THE UNIVERSITY OF MELBOURNE

COMP90038 Algorithms and Complexity

Lecture 4: Analysis of Algorithms

Olya Ohrimenko
(with thanks to Harald Søndergaard)

Announcements

Meet student representatives (see Welcome Module on Canvas for details)

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Last Time: Time Complexity

- Measure **input size** by natural number *n*

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Last Time: Time Complexity

- Measure **input size** by natural number *n*
- Measure **execution time** as number of **basic operations** performed

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Last Time: Time Complexity

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- Mea
  ope

Problem	Size Measure	Basic Operation
Search in a list of *n* items	*n*	Key comparison
Multiply two matrices of floats	Matrix size (rows x columns)	Float multiplication
Compute a^n	log *n*	Float multiplication
Graph problem	Number of nodes and edges	Visiting a node

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- How to **compare** different *t(n) ?*
  - Asymptotic growth rate

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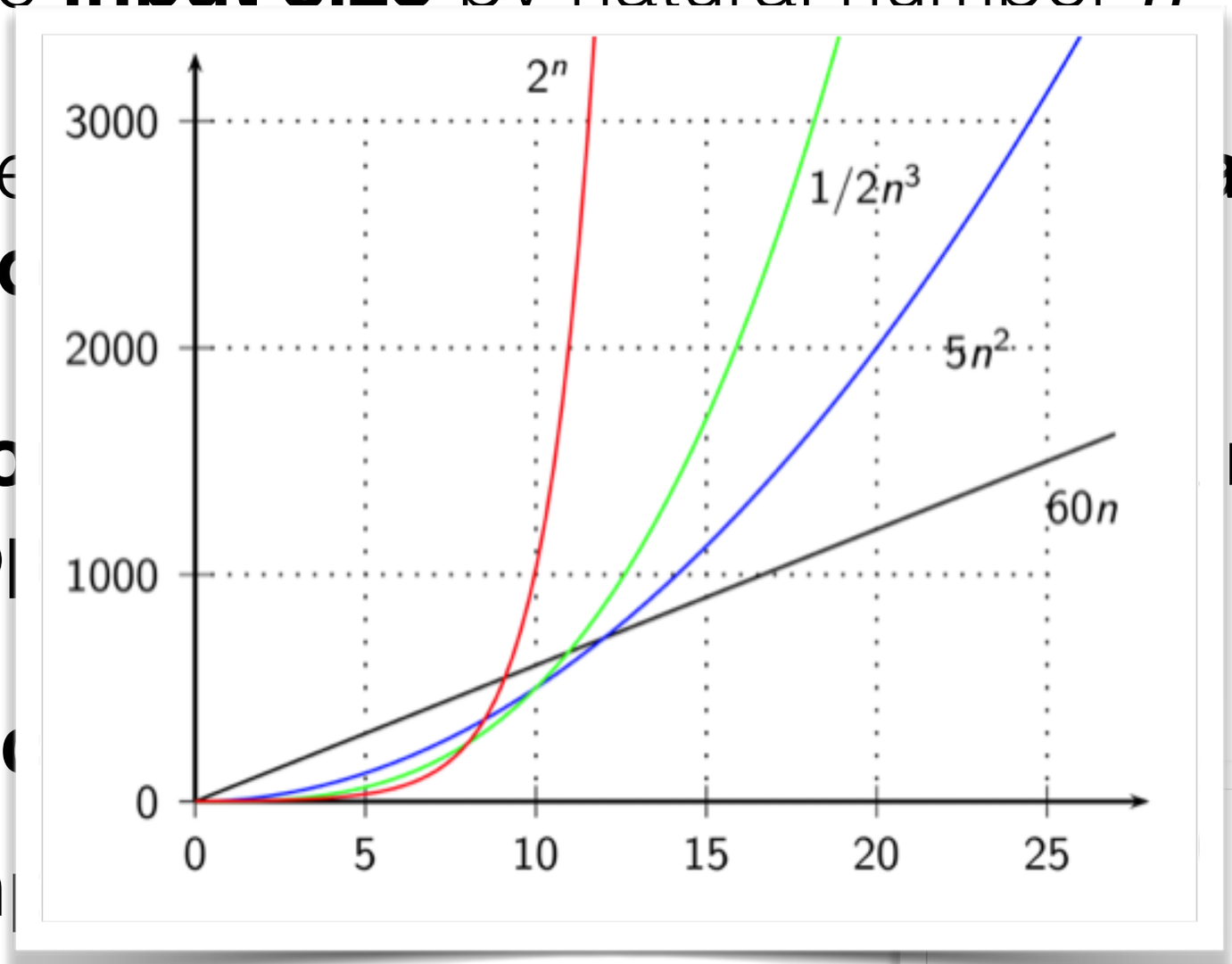
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$$f(n) < g(n) \text{ iff } \lim_{n \to \infty} \frac{f(n)}{g(n)} = 0$$

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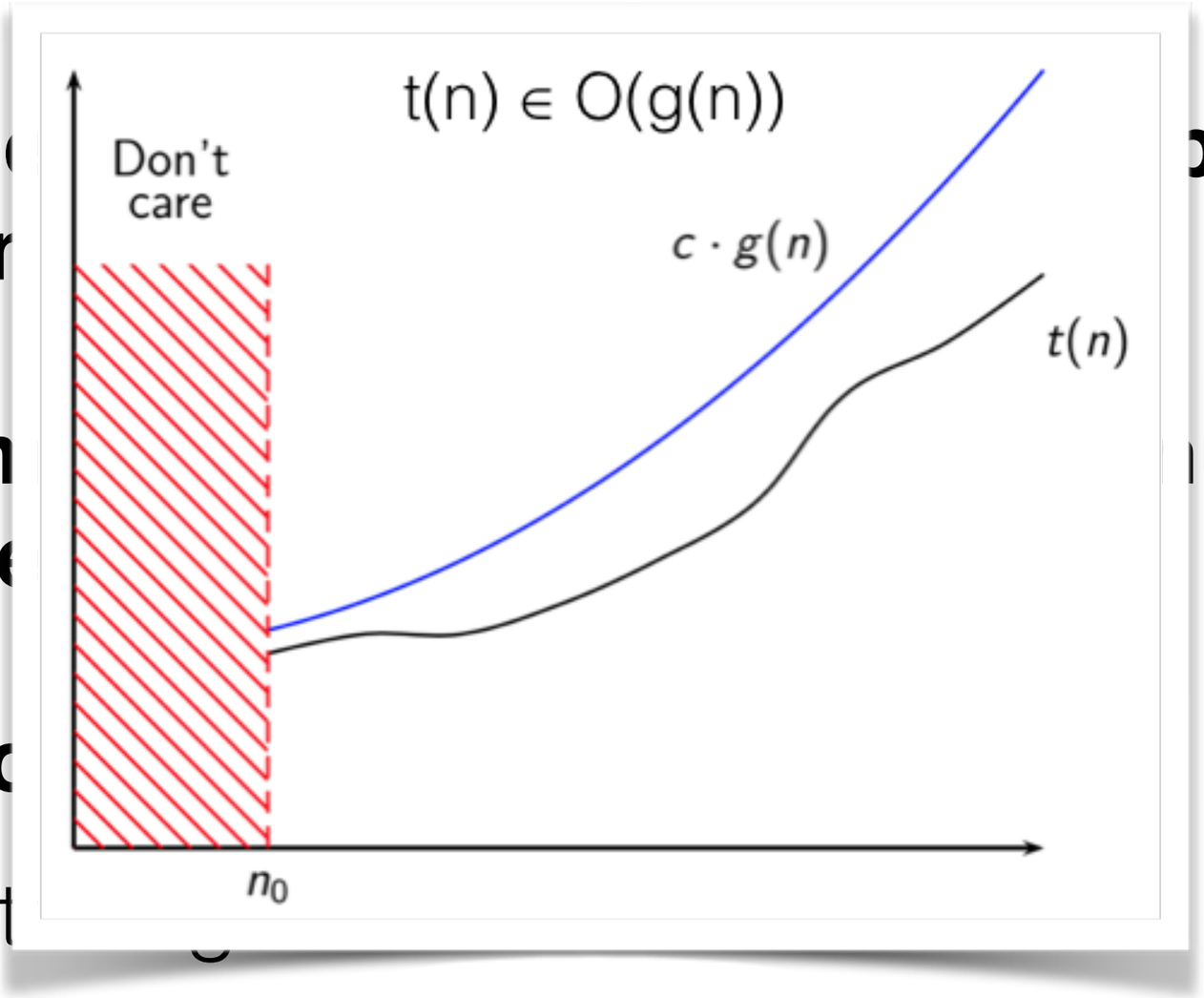
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Last Time: Time Complexity

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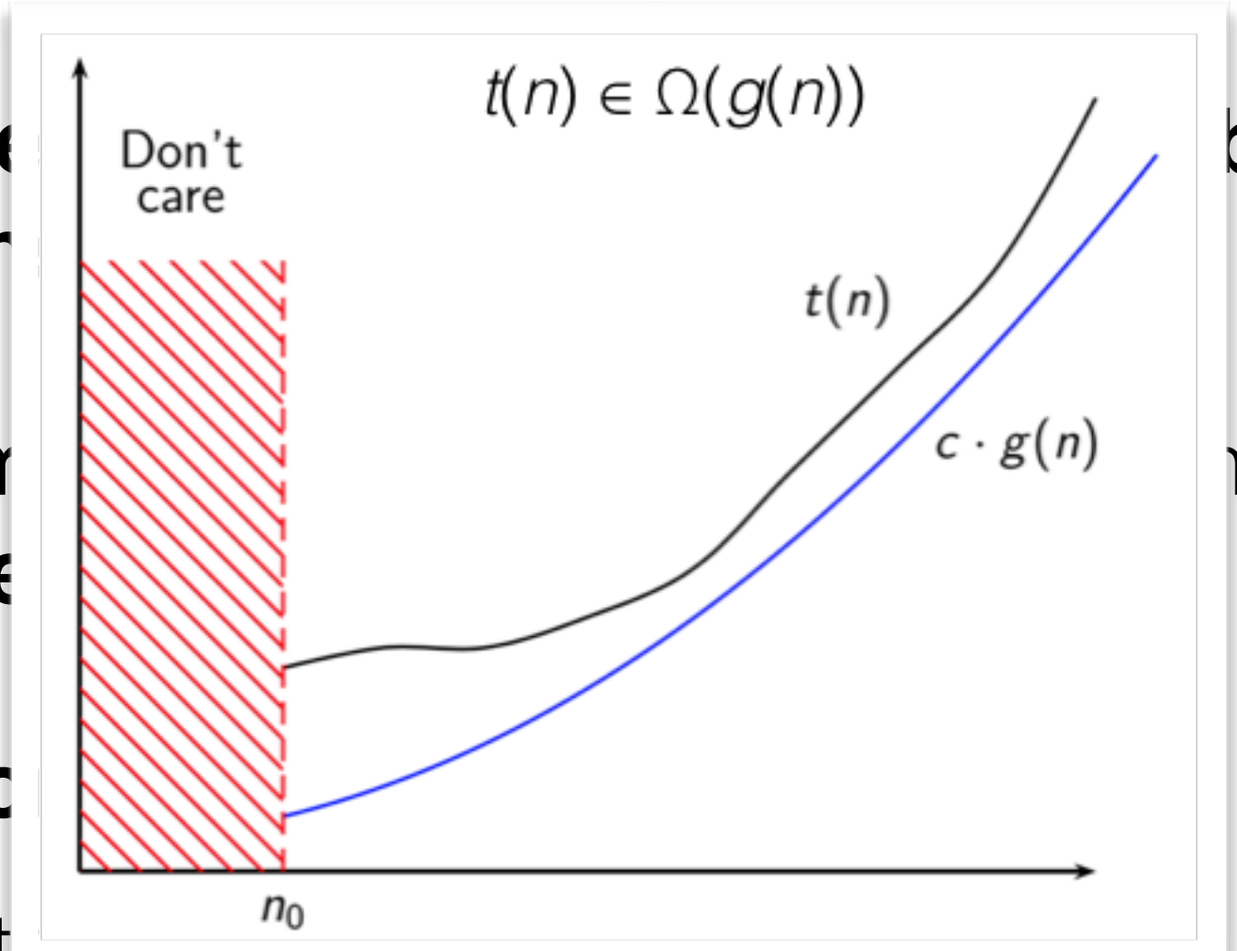
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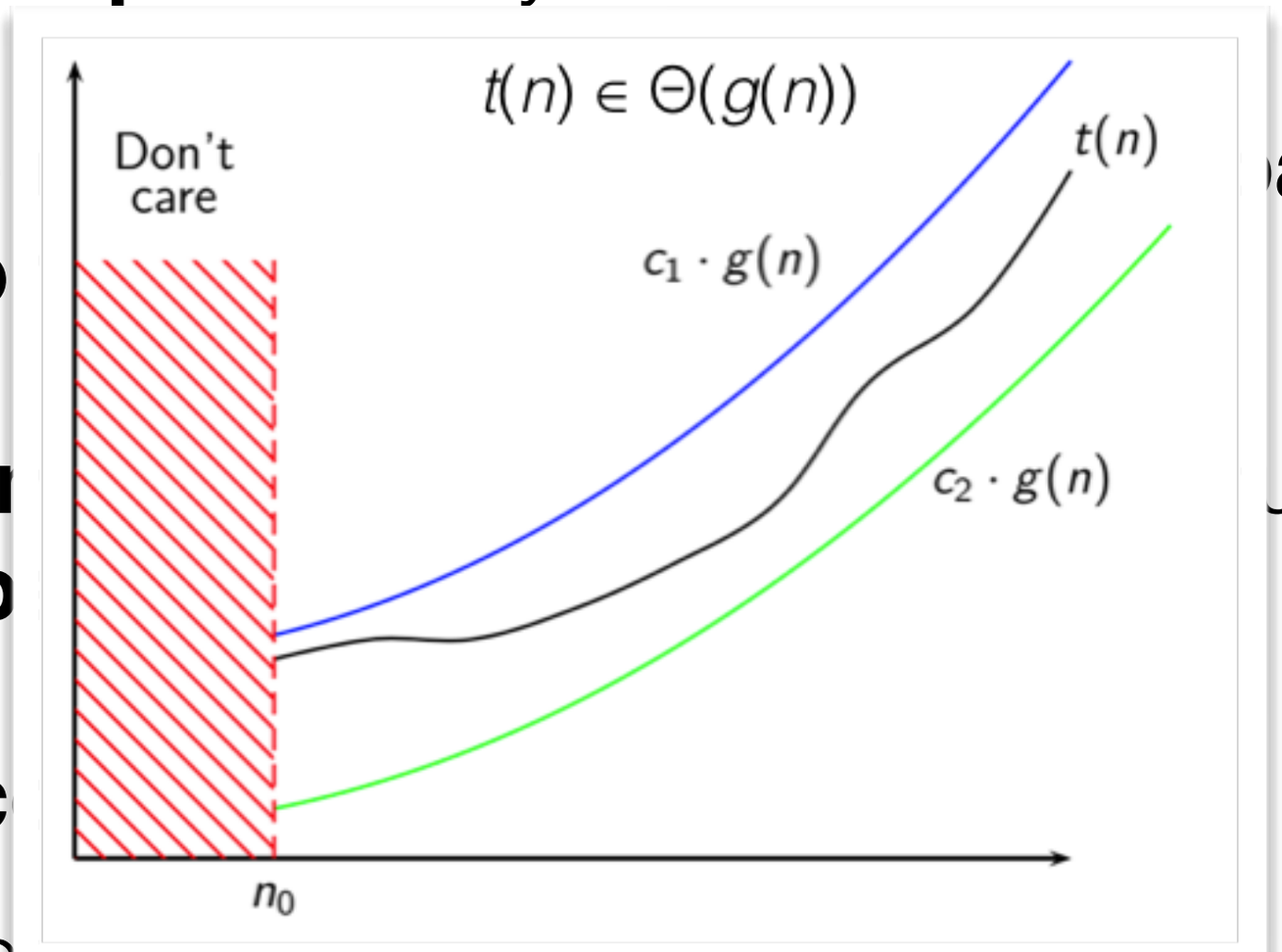
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Establishing Growth Rate

- We proved $t(n) \in O(g(n))$ for some cases of t and g, using the definition of O directly:

  $n > n_0 \Rightarrow t(n) < c \cdot g(n)$ for some c and n_0.

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- $$\lim_{n \to \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & f \text{ grows asymptotically slower than } g \\ c & f \text{ and } g \text{ have same order of growth} \\ \infty & f \text{ grows asymptotically faster than } g \end{cases}$$

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O, Ω, Θ

What this tells us about how $f(n)$ relates to $O(g(n))$, $\Omega(g(n))$ and $\Theta(g(n))$

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$f(n) \in O(g(n))$

f and g have same order of growth

f grows asymptotically faster than g

$f(n) \in \Omega(g(n))$

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Show that $1000n \in O(n^2)$

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L'Hôpital's Rule

$$\lim_{h \to \infty} \frac{t(n)}{g(n)} = \lim_{h \to \infty} \frac{t'(n)}{g'(n)}$$

where t' and g' are the derivatives of t and g

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$$\lim_{n\to\infty} \frac{\log_2 n}{\sqrt{n}} = \lim_{n\to\infty} \frac{\frac{1}{(\log_e 2)n}}{\frac{1}{2\sqrt{n}}}$$

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Example:
Finding Largest Element in an Array

```
function MaxElement(A[·], n)
    max ← A[0]
    for i ← 1 to n − 1 do
        if A[i] > max then
            max ← A[i]
    return max
```

(where ***n*** is length of the array)

A:

23	12	42	6	69	18	3
0	1	2	3	4	5	6

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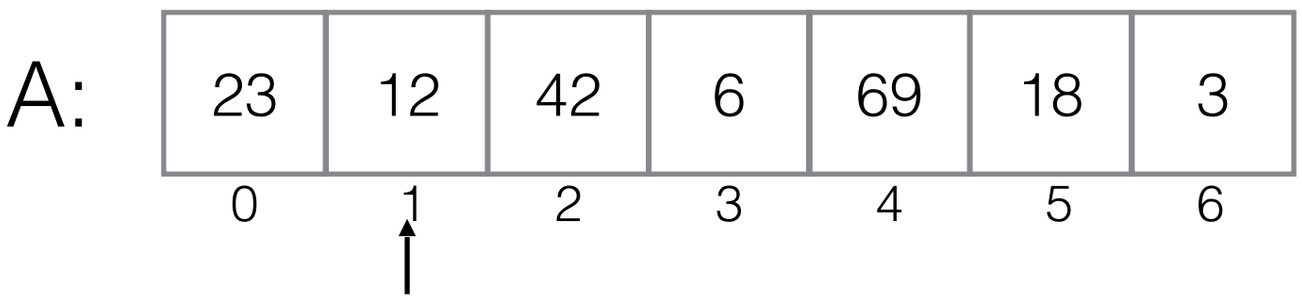
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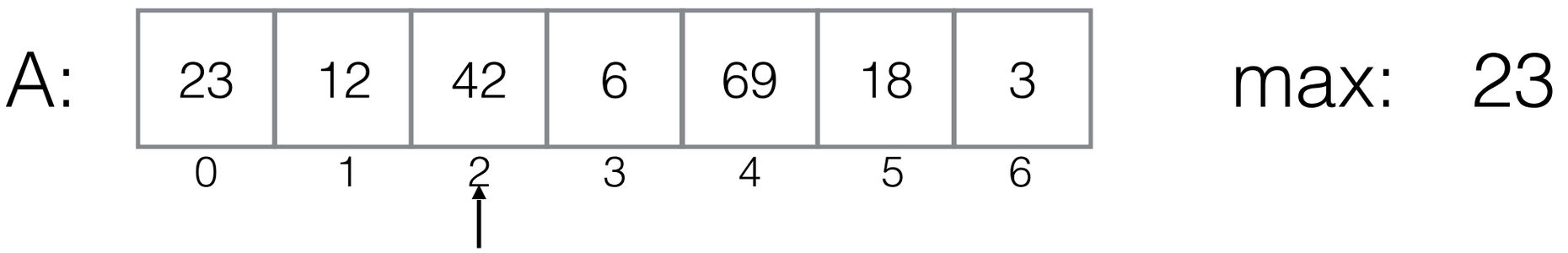
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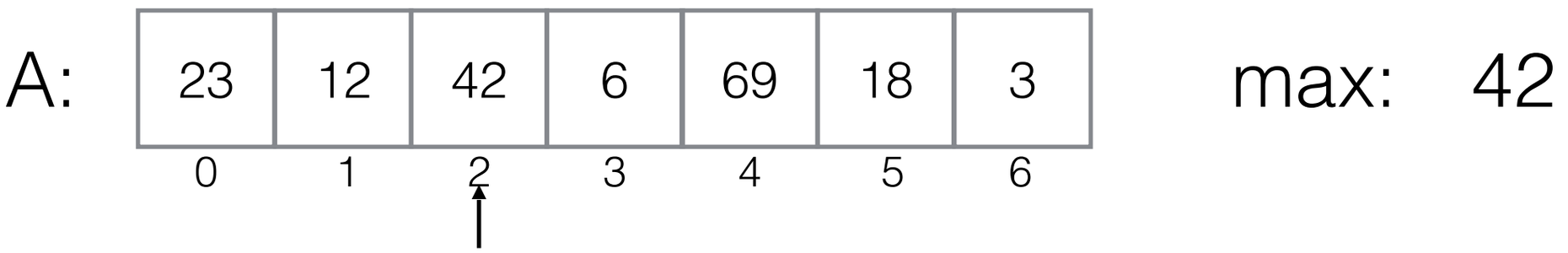
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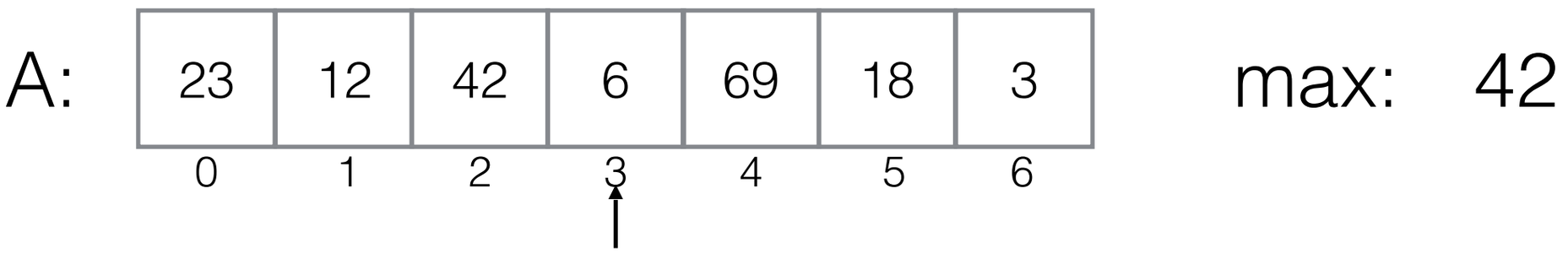


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max: 42

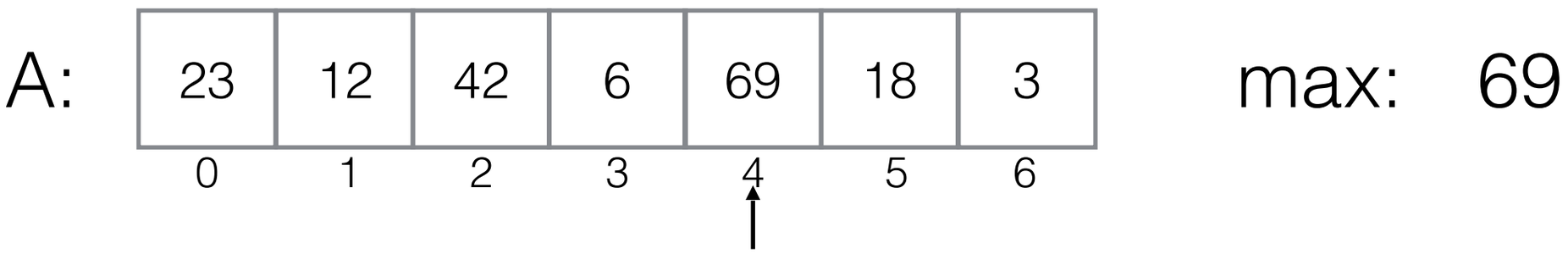
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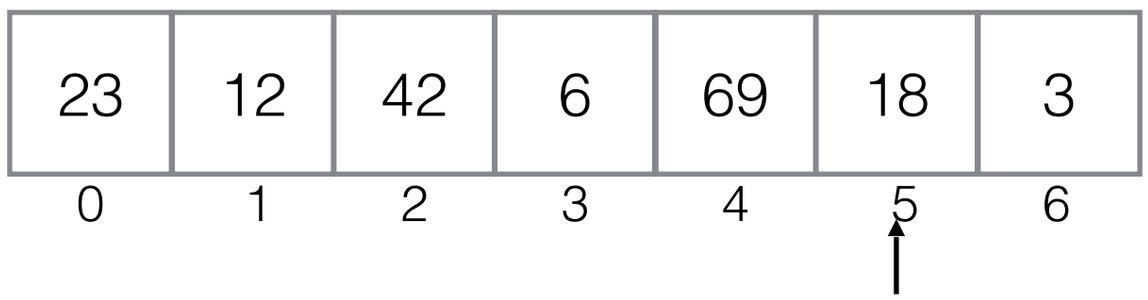
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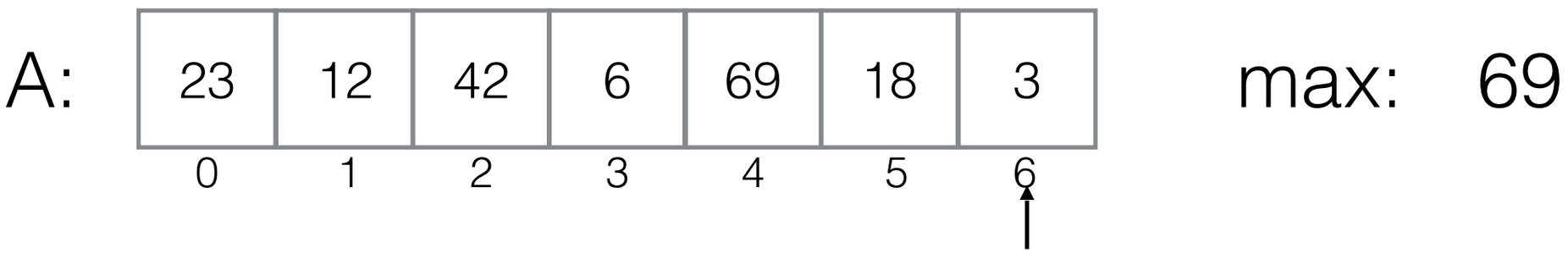
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Example:
Finding Largest Element in an Array

```
function MaxElement(A[·], n)
    max ← A[0]
    for i ← 1 to n − 1 do
        if A[i] > max then
            max ← A[i]
    return max
```

(where ***n*** is length of the array)

Size of input, ***n***:
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(where ***n*** is length of the array)

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Basic operation: c

$$\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$$

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(where ***n*** is length of the array)

Size of input, ***n***:
length of the array

Basic operation: c

Count the number of basic
for an array of size ***n:***

$$\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u - l + 1$$

$$C(n) = \sum_{i=1}^{n-1} 1 = ((n-1) - 1 + 1) = n - 1 \in \Theta(n)$$

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Example: Selection Sort

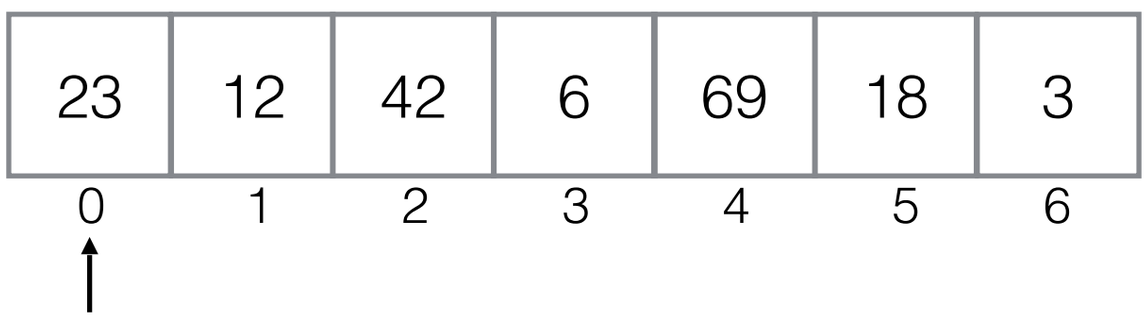
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function SelSort(A[·], n)
    for i ← 0 to n − 2 do
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            if A[j] < A[min] then
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```

23	12	42	6	69	18	3
0	1	2	3	4	5	6

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Example: Selection Sort

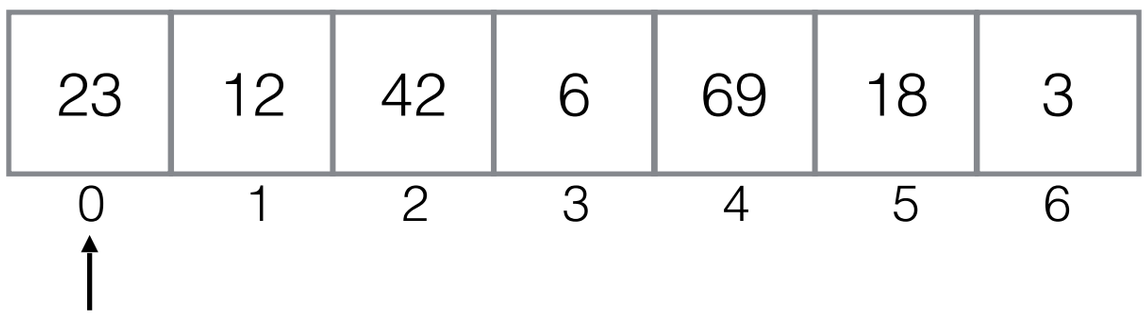
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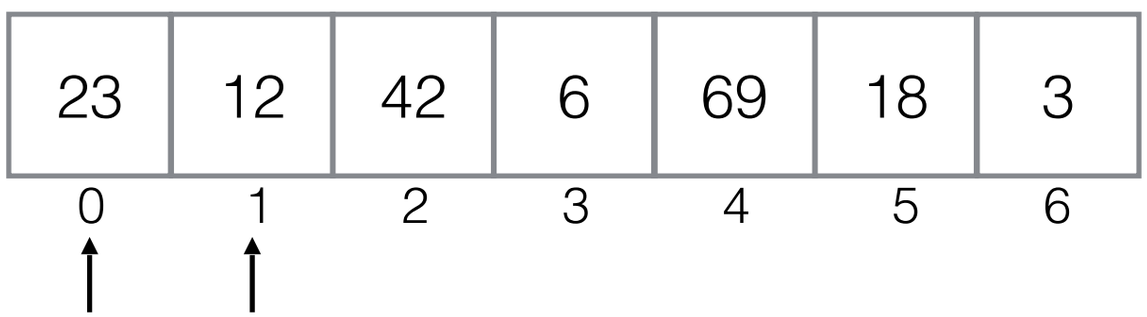
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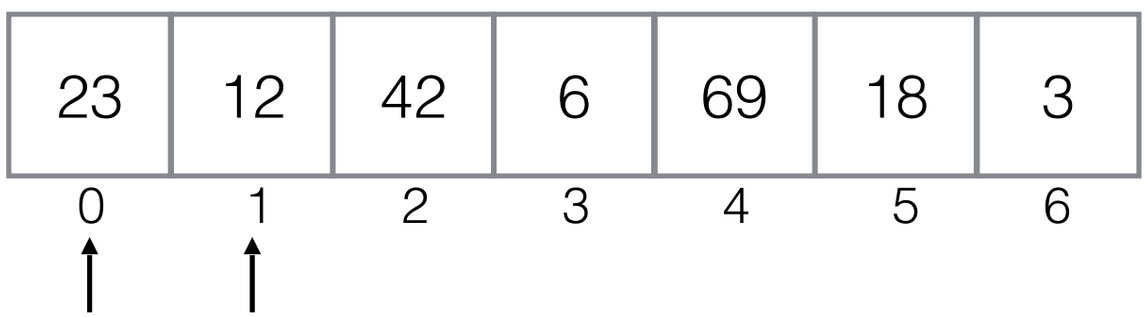
min: 0

A[i] A[j]

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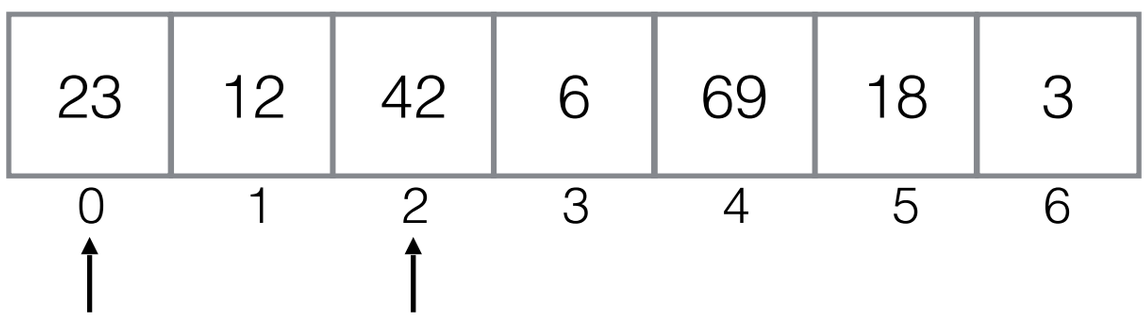


min: 1

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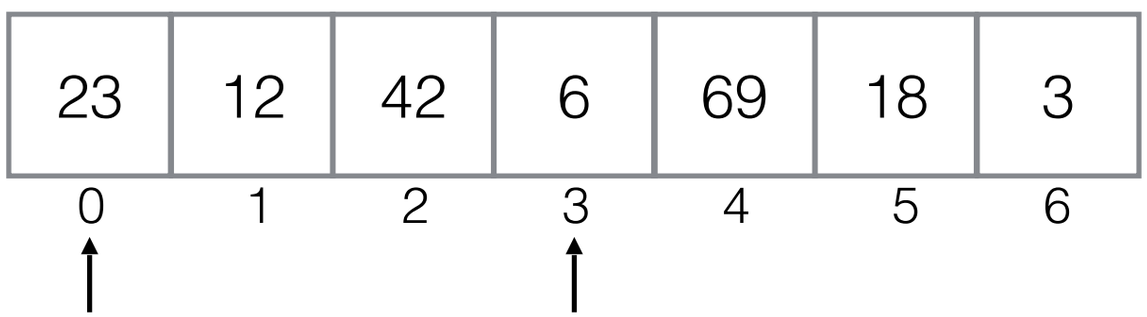


min: 1

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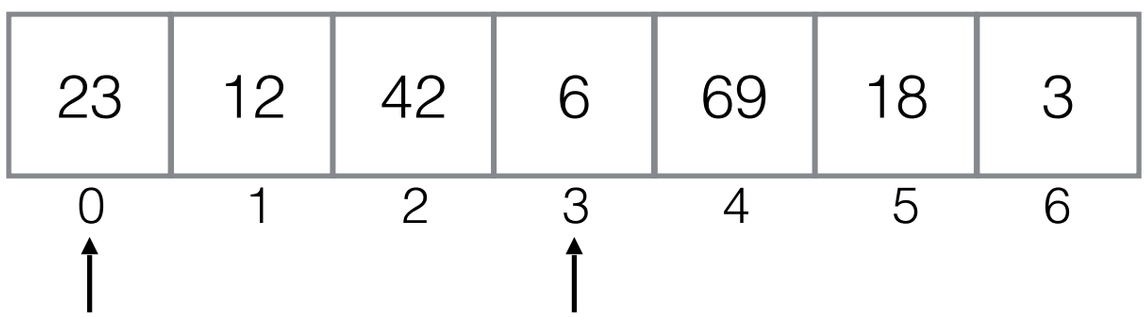


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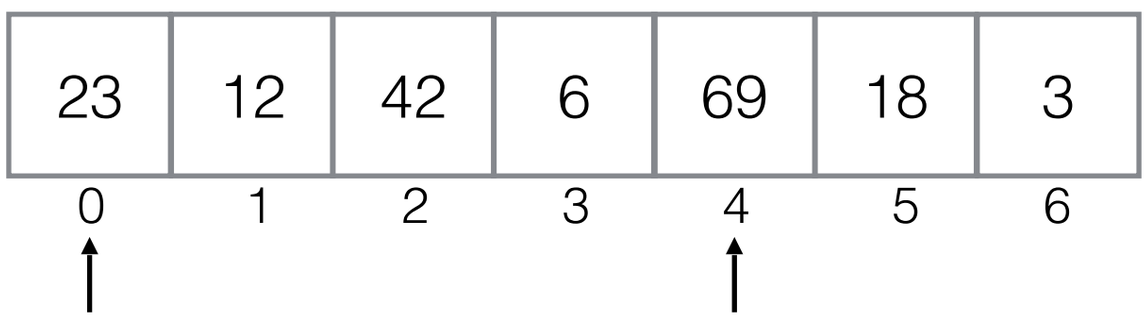


min: 3

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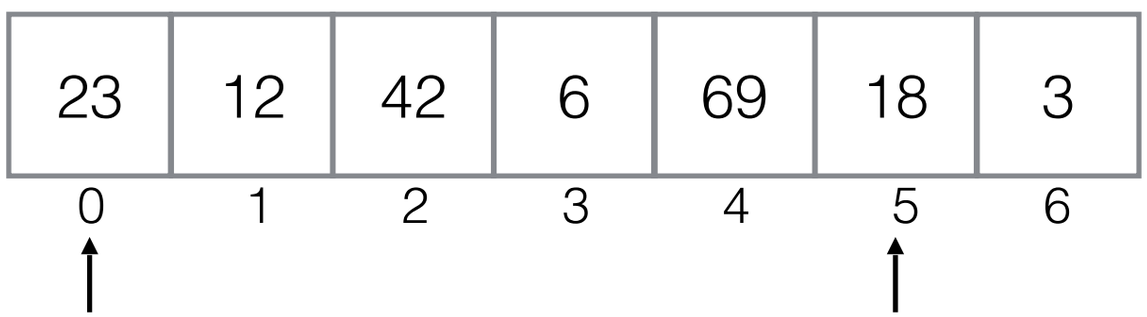


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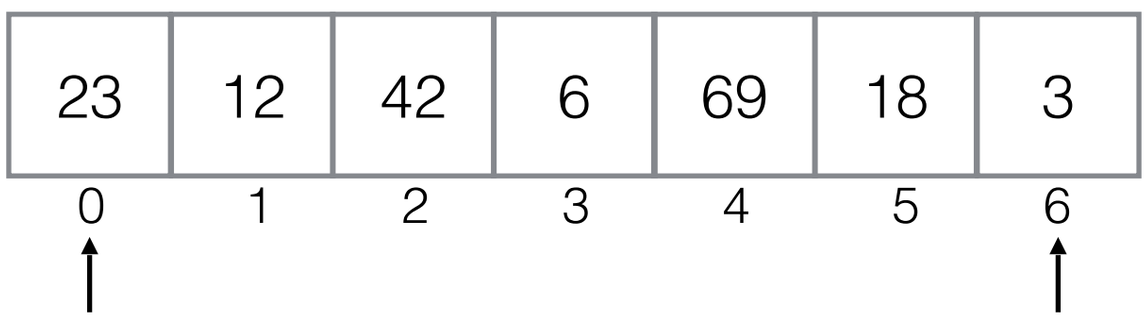


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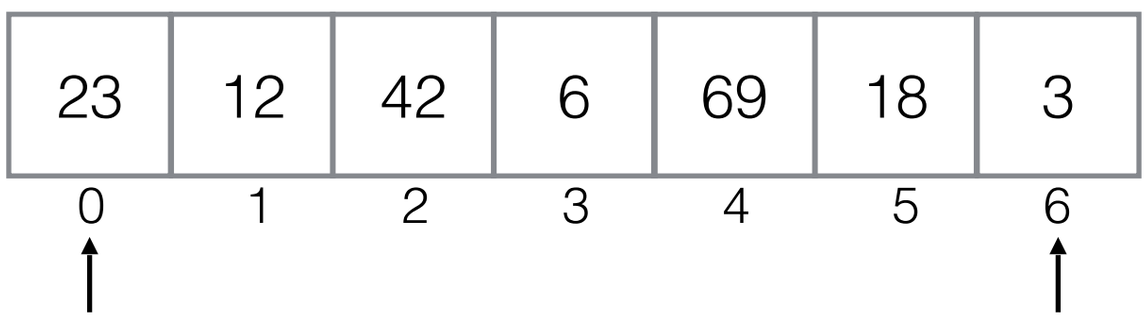


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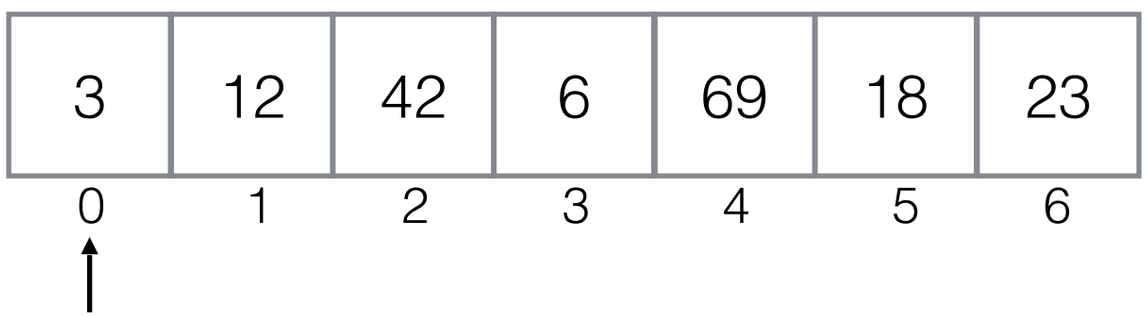


min: 6

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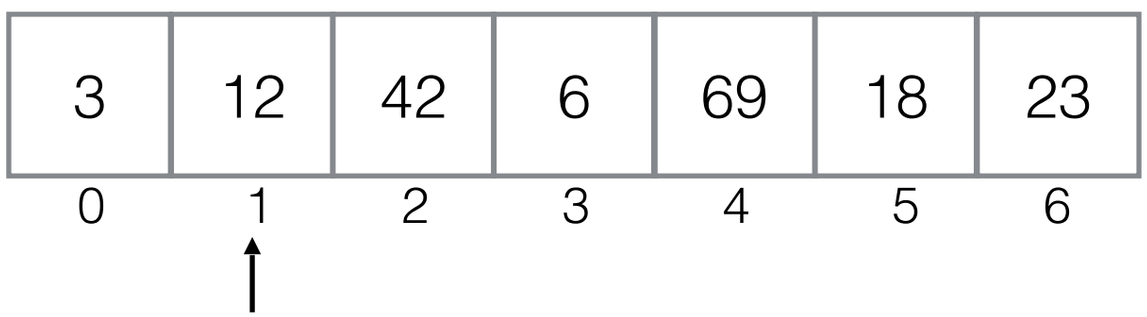


min: 6

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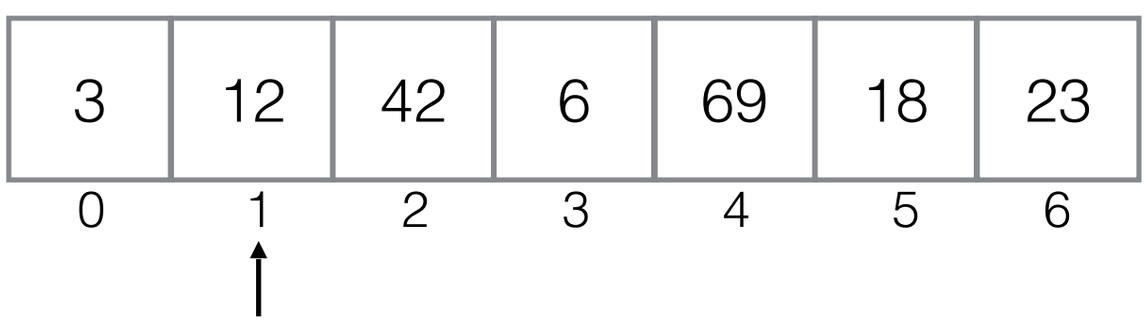


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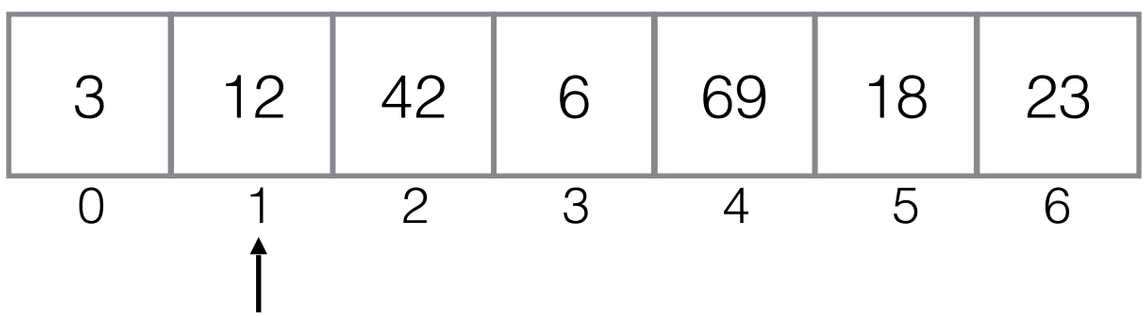
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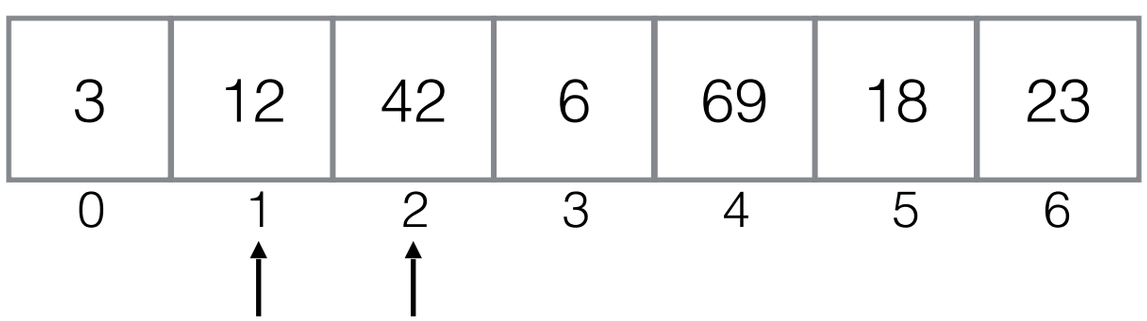


min: 1

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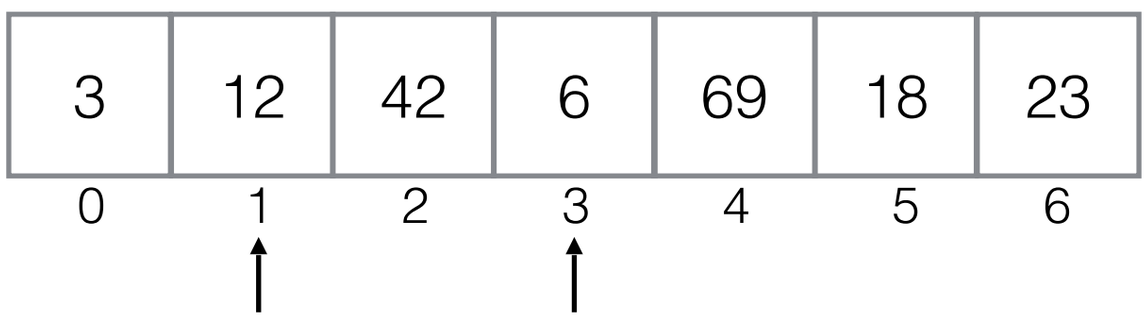
min: 1

A[j] A[j]

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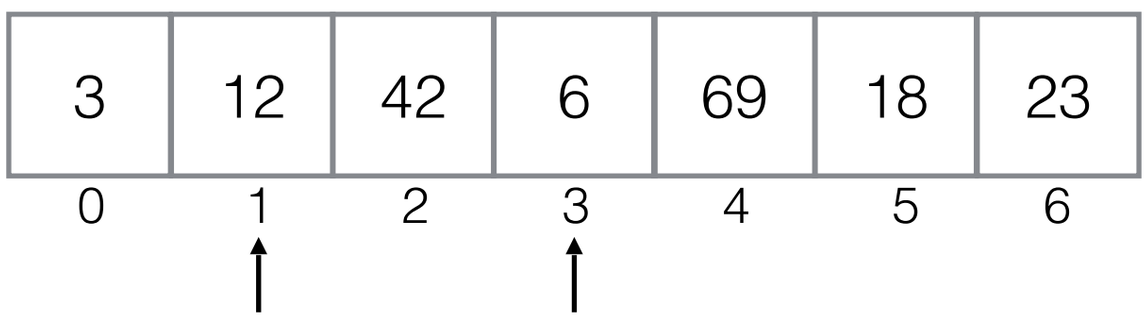


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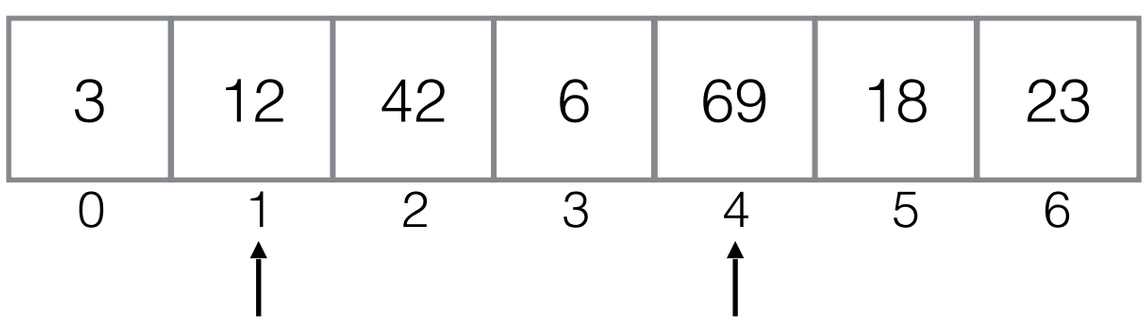


min: 3

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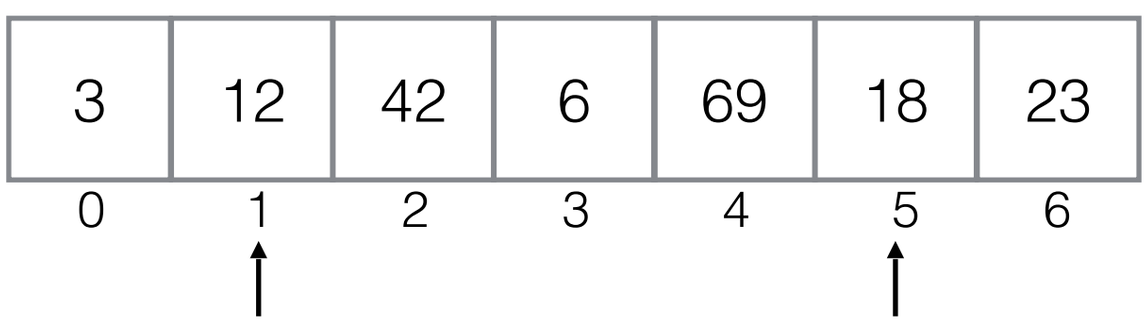


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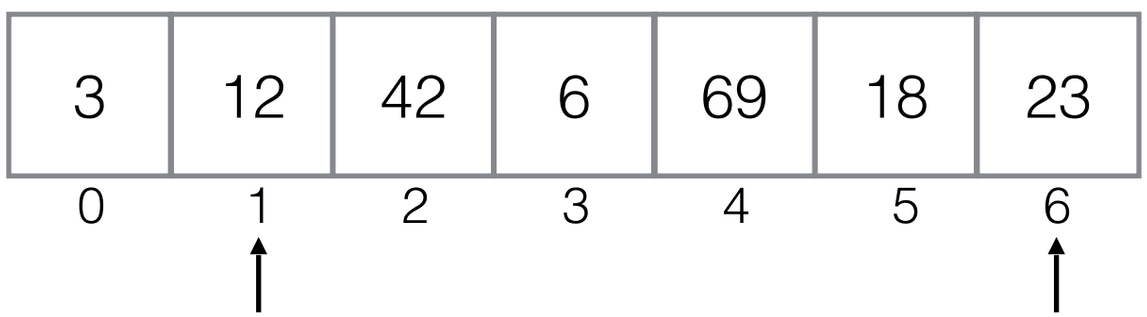


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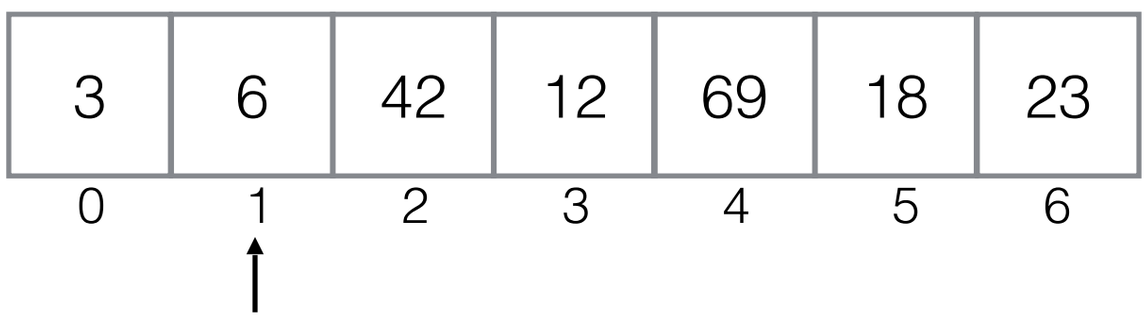


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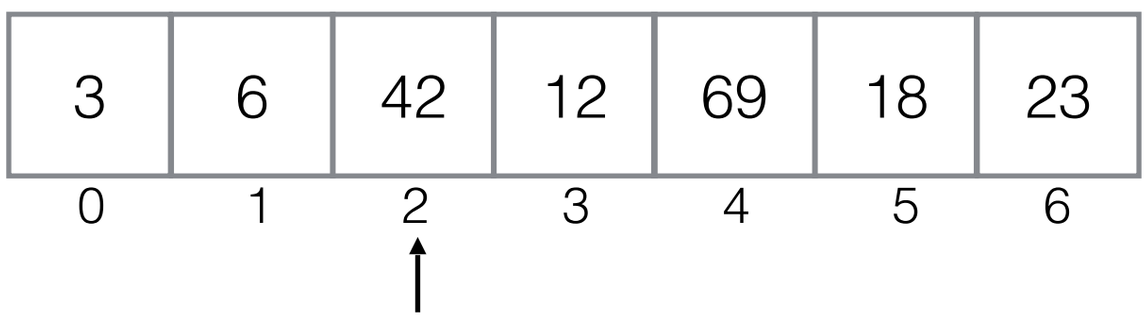


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Input size *n*: length of the array

Basic operation: comparison A[j] < A[min]

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1$$

Using $\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$

$$\sum_{j=i+1}^{n-1} 1$$

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$$\sum_{j=i+1}^{n-1} 1 = (n - 1 - (i + 1) + 1) \quad = (n - 1 \ - i)$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i)$$

Using $\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$

$$\sum_{j=i+1}^{n-1} 1 = (n-1-(i+1)+1) = (n-1-i)$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i)$$

Using

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i)$$

Using $$\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

Using $$\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

Using $\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$

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Using $\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$

$$\sum_{i=0}^{n-2} (n-1) =$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

Using $\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 =$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

Using $\sum_{i=l}^{u} 1 = \underbrace{1+1+\cdots+1}_{u-l+1 \text{ times}} = u-l+1$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 = (n-1)(n-2-0+1) = (n-1)^2$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) \quad = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 = (n-1)(n-2-0+1) = (n-1)^2$$

$$\sum_{i=0}^{n-2} i =$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \approx \frac{1}{2}n^2$$

$$\sum_{i=0}^{n-2} i =$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \approx \frac{1}{2}n^2$$

$$\sum_{i=0}^{n-2} i = \sum_{i=1}^{n-2} i$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \approx \frac{1}{2}n^2$$

$$\sum_{i=0}^{n-2} i = \sum_{i=1}^{n-2} i = \frac{(n-2)(n-2+1)}{2} = \frac{(n-2)(n-1)}{2}$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 = (n-1)(n-2-0+1) = (n-1)^2$$

$$\sum_{i=0}^{n-2} i = \sum_{i=1}^{n-2} i = \frac{(n-2)(n-2+1)}{2} = \frac{(n-2)(n-1)}{2}$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 = (n-1)(n-2-0+1) = (n-1)^2$$

$$\sum_{i=0}^{n-2} i = \sum_{i=1}^{n-2} i = \frac{(n-2)(n-2+1)}{2} = \frac{(n-2)(n-1)}{2}$$

$$C(n) = (n-1)^2 - \frac{(n-2)(n-1)}{2}$$

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Selection Sort Analysis

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i) = \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i$$

$$\sum_{i=0}^{n-2} (n-1) = (n-1) \sum_{i=0}^{n-2} 1 = (n-1)(n-2-0+1) = (n-1)^2$$

$$\sum_{i=0}^{n-2} i = \sum_{i=1}^{n-2} i = \frac{(n-2)(n-2+1)}{2} = \frac{(n-2)(n-1)}{2} = \frac{n(n-1)}{2} - \frac{(n-1)}{2}$$

$$C(n) = (n-1)^2 - \frac{(n-2)(n-1)}{2} \in \Theta(n^2)$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
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      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 0

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 0

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 0 & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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```
function MatrixMult(A[·, ·], B[·, ·], n)        ▷ For n × n matrices
    for i ← 0 to n − 1 do
        for j ← 0 to n − 1 do
            C[i, j] ← 0.0
            for k ← 0 to n − 1 do
                C[i, j] ← C[i, j] + A[i, k] · B[k, j]
    return C
```

i: 0

j: 0

k: 0

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 0 & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 0

k: 0

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} 40 & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 0

k: 1

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 40 & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 0

k: 1

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} 103 & \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} 103 & \\ & \end{bmatrix}$$

A B C

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 103 & 0 \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

k: 0

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 103 & 0 \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

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function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

k: 0

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} 103 & 10 \\ & \end{bmatrix}$$

A B C

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

k: 1

$$A = \begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 103 & 10 \\ & \end{bmatrix}$$

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Example: Matrix Multiplication

function MATRIXMULT($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

i: 0

j: 1

k: 1

$$\begin{bmatrix} 5 & 7 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 \\ 9 & 6 \end{bmatrix} \quad \begin{bmatrix} 103 & 52 \\ & \end{bmatrix}$$

A B C

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Example: Matrix Multiplication

function MatrixMult($A[\cdot,\cdot], B[\cdot,\cdot], n$) ▷ For $n \times n$ matrices
  for $i \leftarrow 0$ to $n-1$ **do**
    for $j \leftarrow 0$ to $n-1$ **do**
      $C[i,j] \leftarrow 0.0$
      for $k \leftarrow 0$ to $n-1$ **do**
        $C[i,j] \leftarrow C[i,j] + A[i,k] \cdot B[k,j]$
  return C

Basic operation: multiplication A[i,k] * B[k,j]

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1)$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

$$\sum_{i=l}^{u} c a_i = c \sum_{i=l}^{u} a_i$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

$$= \sum_{i=0}^{n-1} \left(n \cdot \sum_{j=0}^{n-1} 1 \right)$$

$$\sum_{i=l}^{u} c a_i = c \sum_{i=l}^{u} a_i$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

$$= \sum_{i=0}^{n-1} \left(n \cdot \sum_{j=0}^{n-1} 1 \right) = \sum_{i=0}^{n-1} n^2$$

$$\sum_{i=l}^{u} c a_i = c \sum_{i=l}^{u} a_i$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

$$= \sum_{i=0}^{n-1} \left(n \cdot \sum_{j=0}^{n-1} 1 \right) = \sum_{i=0}^{n-1} n^2$$

$$= n^3$$

$$\sum_{i=l}^{u} c a_i = c \sum_{i=l}^{u} a_i$$

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Example: Matrix Multiplication

$$M(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1$$

$$\sum_{i=l}^{u} 1 = \underbrace{1 + 1 + \cdots + 1}_{u-l+1 \text{ times}} = u - l + 1$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} ((n-1) - 0 + 1) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (n \cdot 1)$$

$$= \sum_{i=0}^{n-1} \left(n \cdot \sum_{j=0}^{n-1} 1 \right) = \sum_{i=0}^{n-1} n^2$$

$$\sum_{i=l}^{u} ca_i = c \sum_{i=l}^{u} a_i$$

$$= n^3$$

$$\in \Theta(n^3)$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$F(5)$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$F(5) = F(4) \cdot 5$$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5
\end{aligned}
$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5 \\
&= ((F(2) \cdot 3) \cdot 4) \cdot 5
\end{aligned}
$$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5 \\
&= ((F(2) \cdot 3) \cdot 4) \cdot 5 \\
&= (((F(1) \cdot 2) \cdot 3) \cdot 4) \cdot 5
\end{aligned}
$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5 \\
&= ((F(2) \cdot 3) \cdot 4) \cdot 5 \\
&= (((F(1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= ((((F(0) \cdot 1) \cdot 2) \cdot 3) \cdot 4) \cdot 5
\end{aligned}
$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
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$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5 \\
&= ((F(2) \cdot 3) \cdot 4) \cdot 5 \\
&= (((F(1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= ((((F(0) \cdot 1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= ((((1 \cdot 1) \cdot 2) \cdot 3) \cdot 4) \cdot 5
\end{aligned}
$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

$$
\begin{aligned}
F(5) &= F(4) \cdot 5 \\
&= (F(3) \cdot 4) \cdot 5 \\
&= ((F(2) \cdot 3) \cdot 4) \cdot 5 \\
&= (((F(1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= ((((F(0) \cdot 1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= ((((1 \cdot 1) \cdot 2) \cdot 3) \cdot 4) \cdot 5 \\
&= 5!
\end{aligned}
$$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

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Analysing Recursive Algorithms

function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) =$$

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Analysing Recursive Algorithms

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```
function F(n)
    if n = 0 then return 1
    else return F(n − 1) · n
```

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) = \quad 0$$

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Analysing Recursive Algorithms



function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) = \quad 0$$
$$M(n) =$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) = 0$$
$$M(n) = M(n-1) + 1$$

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Analysing Recursive Algorithms

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function $\mathrm{F}(n)$
    if $n = 0$ **then return** 1
    else return $\mathrm{F}(n-1) \cdot n$

Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) = 0$$
$$M(n) = M(n-1) + 1$$

Need to express M(n) in **closed form** (i.e. non-recursively)

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Analysing Recursive Algorithms

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  if $n = 0$ **then return** 1
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Basic operation: multiplication

We express the cost **recursively** (as a **recurrence relation**)

$$M(0) = 0$$
$$M(n) = M(n-1) + 1$$

Need to express M(n) in **closed form** (i.e. non-recursively)

Try: **“telescoping”** aka **“backward substitution”**

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ? $M(n-1) = M((n-1)-1) + 1$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) =$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

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$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$

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Telescoping
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$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$

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Telescoping
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What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$

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Telescoping
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$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

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$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$
$$= n$$

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Telescoping
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What is M(n-1) ?

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$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$
$$= n$$

Closed form:

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$
$$= n$$

Closed form:

$$M(n) = n$$

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Telescoping
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$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
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$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$
$$= n$$

Closed form:

$$M(n) = n$$

Complexity:

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Telescoping
(aka Backward Substitution)

$$M(n) = M(n-1) + 1 \qquad M(0) = 0$$

What is M(n-1) ?

$$M(n-1) = M((n-1)-1) + 1$$
$$= M(n-2) + 1$$

$$M(n) = (M(n-2) + 1) + 1$$
$$= M(n-2) + 2$$
$$= (M(n-3) + 1) + 2$$
$$= M(n-3) + 3$$
$$\cdots$$
$$= M(n-n) + n$$
$$= M(0) + n$$
$$= n$$

Closed form:

$$M(n) = n$$

Complexity:

$$M(n) \in \Theta(n)$$

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Example:
Binary Search in Sorted Array

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```
function BinSearch(A[·], lo, hi, key)
    if lo > hi then return −1
    mid ← lo + (hi − lo)/2
    if A[mid] = key then return mid
    else
        if A[mid] > key then
            return BinSearch(A, lo, mid − 1, key)
        else return BinSearch(A, mid + 1, hi, key)
```

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

```
function BinSearch(A[·], lo, hi, key)
    if lo > hi then return −1
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```

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

function BINSEARCH($A[\cdot], lo, hi, key$)
    if $lo > hi$ **then return** -1
    $mid \leftarrow lo + (hi - lo)/2$
    if $A[mid] = key$ **then return** mid
    else
        if $A[mid] > key$ **then**
            return BINSEARCH($A, lo, mid - 1, key$)
        else return BINSEARCH($A, mid + 1, hi, key$)

lo: 0

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

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Example:
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    else
        if A[mid] > key then
            return BinSearch(A, lo, mid − 1, key)
    else return BinSearch(A, mid + 1, hi, key)
```

lo: 0

hi: 6

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

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Example:
Binary Search in Sorted Array

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    else
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            return BinSearch(A, lo, mid − 1, key)
    else return BinSearch(A, mid + 1, hi, key)
```

lo: 0

hi: 6

key: 41

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

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Example:
Binary Search in Sorted Array

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    else
        if A[mid] > key then
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    else return BinSearch(A, mid + 1, hi, key)
```

lo: 0

hi: 6

key: 41

mid: 3

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

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```

lo: 0

hi: 6

key: 41

mid: 3

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,0,6,41)

BinSearch(A,4,6,41)

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

```
function BinSearch(A[·], lo, hi, key)
    if lo > hi then return −1
    mid ← lo + (hi − lo)/2
    if A[mid] = key then return mid
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```

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,4,6,41)

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Example:
Binary Search in Sorted Array

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BinSearch(A,4,6,41)

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Binary Search in Sorted Array

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```

lo: 4

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,4,6,41)

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Example:
Binary Search in Sorted Array

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    else return BinSearch(A, mid + 1, hi, key)
```

lo: 4

hi: 6

key: 41

mid: 5

A:

0	1	2	3	4	5	6
4	9	13	22	41	83	96

BinSearch(A,4,6,41)

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

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```

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0	1	2	3	4	5	6

BinSearch(A,4,6,41)

BinSearch(A,4,4,41)

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Example:
Binary Search in Sorted Array

THE UNIVERSITY OF MELBOURNE

```
function BinSearch(A[·], lo, hi, key)
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```

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0	1	2	3	4	5	6

BinSearch(A,4,4,41)

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Example:
Binary Search in Sorted Array

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```

A:

4	9	13	22	41	83	96
0	1	2	3	4	5	6

BinSearch(A,4,4,41)

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Example:
Binary Search in Sorted Array

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function BinSearch(A[·], lo, hi, key)
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lo: 4

hi: 4

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```

lo: 4

hi: 4

key: 41

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lo: 4

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key: 41

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lo: 4

hi: 4

key: 41

mid: 4

A:

4	9	13	22	41	83	96
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BinSearch(A,4,4,41)

returns 4

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$$C(0) = 0$$

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Basic operation: key comparison A[mid] = key

$$C(0) = 0 \qquad C(n) = C(n/2) + 1$$

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Telescoping

A **smoothness rule** allows us to assume that n is a power of 2

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$$C(n) = C(n/2) + 1$$
$$= (C(n/4) + 1) + 1$$

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Telescoping

A **smoothness rule** allows us to assume that n is a power of 2

$$C(0) = 0 \qquad C(n) = C(n/2) + 1$$

$$\begin{aligned} C(n) &= C(n/2) + 1 \\ &= (C(n/4) + 1) + 1 \\ &= ((C(n/8) + 1) + 1) + 1 \end{aligned}$$

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$$C(n) \in \Theta(\log n)$$

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Logarithmic Functions Have Same Rate of Growth

In O, Ω, Θ, expressions we can just write “log” for any logarithmic function no matter what the base is

Asymptotically, all logarithmic behaviour is the same, since

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$$\lim_{n\to\infty} \frac{log_a n}{log_b n} = \lim_{n\to\infty} \frac{(\log_a b)(\log_b n)}{\log_b n} = (\log_a b) \lim_{n\to\infty} 1 = \log_a b$$

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So, e.g.: $\Theta(log_a n) = \Theta(log_b n)$

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So, e.g.: $\Theta(log_a n) = \Theta(log_b n)$

Also, since $\log n^c = c \cdot \log n$ $\log n^c \in \Theta(\log n)$

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Summarising Reasoning with Big-Oh

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Summarising Reasoning with Big-Oh

Suppose

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Summarising Reasoning with Big-Oh

Suppose $t_1(n) \in O(g_1(n))$ $t_2(n) \in O(g_2(n))$

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Summarising Reasoning with Big-Oh

Suppose $t_1(n) \in O(g_1(n))$ $t_2(n) \in O(g_2(n))$

Then the following hold:

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Summarising Reasoning with Big-Oh

Suppose $t_1(n) \in O(g_1(n))$ $t_2(n) \in O(g_2(n))$

Then the following hold:

$$t_1(n) + t_2(n) \in O(max\{g_1(n), g_2(n)\})$$

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(we can throw away smaller summands)

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$$t_1(n) \cdot t_2(n) \in O(g_1(n) \cdot g_2(n))$$

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Summarising Reasoning with Big-Oh

Suppose $t_1(n) \in O(g_1(n))$ $t_2(n) \in O(g_2(n))$

Then the following hold:

$$t_1(n) + t_2(n) \in O(max\{g_1(n), g_2(n)\})$$

(we can throw away smaller summands)

$$c \cdot t_1(n) \in O(g_1(n))$$

(constants can be thrown away too)

$$t_1(n) \cdot t_2(n) \in O(g_1(n) \cdot g_2(n))$$

(for nested loops: count number of times outer loop is executed, multiply by cost of inner loop)

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Some Useful Formulas

From Stirling's formula: $n! \in O(n^{n+\frac{1}{2}})$

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From Stirling's formula: $n! \in O(n^{n+\frac{1}{2}})$

this is not the time complexity of an algorithm to compute n-factorial

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Some Useful Formulas

From Stirling's formula: $n! \in O(n^{n+\frac{1}{2}})$

this is not the time complexity of an algorithm to compute n-factorial

$$\sum_{i=0}^{n} i^2 = \frac{n}{3}(n+\tfrac{1}{2})(n+1)$$

$$\sum_{i=0}^{n}(2i+1) = (n+1)^2$$

$$\sum_{i=1}^{n} 1/i = O(\log n)$$

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See also Cormen's Appendix A or Levitin's Appendix A.

Levitin's Appendix B is a tutorial on recurrence relations.

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The Road Ahead

- You'll get much more familiar with asymptotic analysis as we use it on algorithms we meet in this course.

- Next week we begin our study of algorithms by looking at **brute force** approaches

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Reminders

- Weekly quizzes
- Diagnostic quizzes
- Plenty of ways to get help: consultation hours, CIS Student Hub, Ed, see Canvas for details.
- The textbook is available in the library catalogue.
- Attempt tutorial questions before tutorials.

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Back to Euclid

```
function EUCLID(m, n)
    while n ≠ 0 do
        r ← m mod n
        m ← n
        n ← r
    return m
```

```
function EUCLID(m, n)
    if n = 0 then
        return m
    return EUCLID(n, m mod n)
```


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Back to Euclid

function EUCLID(m, n)
  while $n \neq 0$ **do**
    $r \leftarrow m \bmod n$
    $m \leftarrow n$
    $n \leftarrow r$
  return m

function EUCLID(m, n)
  if $n = 0$ **then**
    return m
  return EUCLID$(n, m \bmod n)$

Running time is **linear** in size (in bits) of input, i.e.

$$O(\log(m + n))$$

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since the value of m (and n) is at least halved in every two iterations.

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$$O(\log(m+n))$$

since the value of m (and n) is at least halved in every two iterations.

Why? After two iterations, m becomes m mod n; also

$$1 < n < m \implies m \bmod n < m/2$$

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